Analyzing the Traffic Crash Incidents in Somerville

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1. Abstract 1
2 . Methodology 2
3. P-Diagram 4
4. Data 5
5. Descriptive Statistics: Distributions 6
6. Descriptive Statistics: Box Plot 12
7. Descriptive Statistics: Graph 14
8. Statistical Analysis 20
9. Conclusion 24
10. Suggestions 26
11. Takeaways 27
12. Future Discussions/Possibilities 28
13. References 29

1. Abstract

The increasing complexity of modern transportation systems has made road traffic safety a critical global concern, with millions of lives affected by crashes each year. This study addresses a fundamental aspect of road safety by investigating the influence of environmental factors—specifically weather conditions, light levels, and road surface types—on the number of suspected injuries resulting from traffic crashes. By analysing real-world crash data, this research seeks to uncover patterns and relationships that offer actionable insights into reducing crash severity and improving public safety.

The data utilized in this study consists of meticulously recorded crash incidents, providing detailed information about environmental conditions, crash circumstances, and the corresponding injury counts. Prior to analysis, the dataset was rigorously cleaned to ensure consistency and accuracy, with missing values addressed and key variables categorized for meaningful comparisons. Using descriptive statistics, the data was summarized to highlight distribution patterns and key trends. Inferential statistical techniques, particularly ANOVA, were employed to examine whether significant differences exist in injury counts across categories of weather, light, and road surface conditions.

Visualizations play a pivotal role in this analysis, offering compelling representations of the relationships between environmental factors and number of injuries. Bar charts, comparative tables, and other graphical tools are utilized to create a comprehensive narrative of the findings. These insights are designed not just to inform but to inspire action—guiding policymakers, transportation authorities, and safety advocates in prioritizing interventions for high-risk conditions. Whether through enhanced road maintenance during adverse weather, improved lighting in critical areas, or targeted public awareness campaigns, the study's findings provide a foundation for strategic decision-making.

By leveraging real-world crash data, this research underscores the importance of evidence-based approaches to road safety. While the focus is on weather, light, and road surface conditions, the study also sets the stage for future investigations to incorporate additional variables such as traffic density, vehicle types, and driver behaviour. In doing so, it opens new avenues for understanding and mitigating the multifaceted risks associated with road traffic crashes, fostering a safer and more resilient transportation system for all.

2 . Methodology

This study employs a structured approach to analyse the impact of environmental factors—weather conditions, light conditions, and road surface types—on the number of suspected injuries in traffic crashes. The methodology is divided into the following key stages:

1. Data Acquisition and Preparation

The dataset used for this study was sourced from real-world crash records, containing information on crash dates, locations, environmental conditions, and suspected injury counts. To ensure the dataset's reliability:

- Missing values in critical variables were identified and addressed using imputation or exclusion techniques.
- Inconsistent or redundant entries were removed to maintain data integrity.
- Key variables, such as weather conditions, light conditions, and road surface types, were categorized to facilitate meaningful analysis.

2. Data Exploration and Descriptive Analysis

A comprehensive exploratory data analysis (EDA) was conducted to understand the dataset's characteristics and detect patterns:

- **Distribution Analysis**: The number of suspected injuries was summarized using descriptive statistics, including mean, median, and standard deviation.
- **Visual Representation**: Bar charts and histograms were created to illustrate the distribution of injury counts across various categories.
- **Correlation Assessment**: Preliminary checks were conducted to explore potential relationships between environmental factors and number of injuries.

3. Statistical Analysis

To evaluate the significance of the relationships between environmental factors and the number of suspected injuries, inferential statistical techniques were applied:

- **Analysis of Variance (ANOVA)**: This technique was used to determine whether there are statistically significant differences in the mean number of injuries across categories of weather conditions, light conditions, and road surface types. Each variable was analyzed independently to isolate its impact.
- **Assumption Validation**: Assumptions for ANOVA, including normality and homogeneity of variance, were tested to ensure the validity of the results.

Transformations were applied to the data where necessary to meet these assumptions.

4. Visualization and Reporting

Visualizations were employed to enhance the interpretation of the analysis:

- **Bar Charts**: Used to compare average injury counts across different categories.
- **Summary Tables**: Tabular formats were created to present key statistics and ANOVA results clearly.
- **Trends and Patterns**: Insights were highlighted to emphasize notable differences or lack thereof across the studied variables.

5. Interpretation and Recommendations

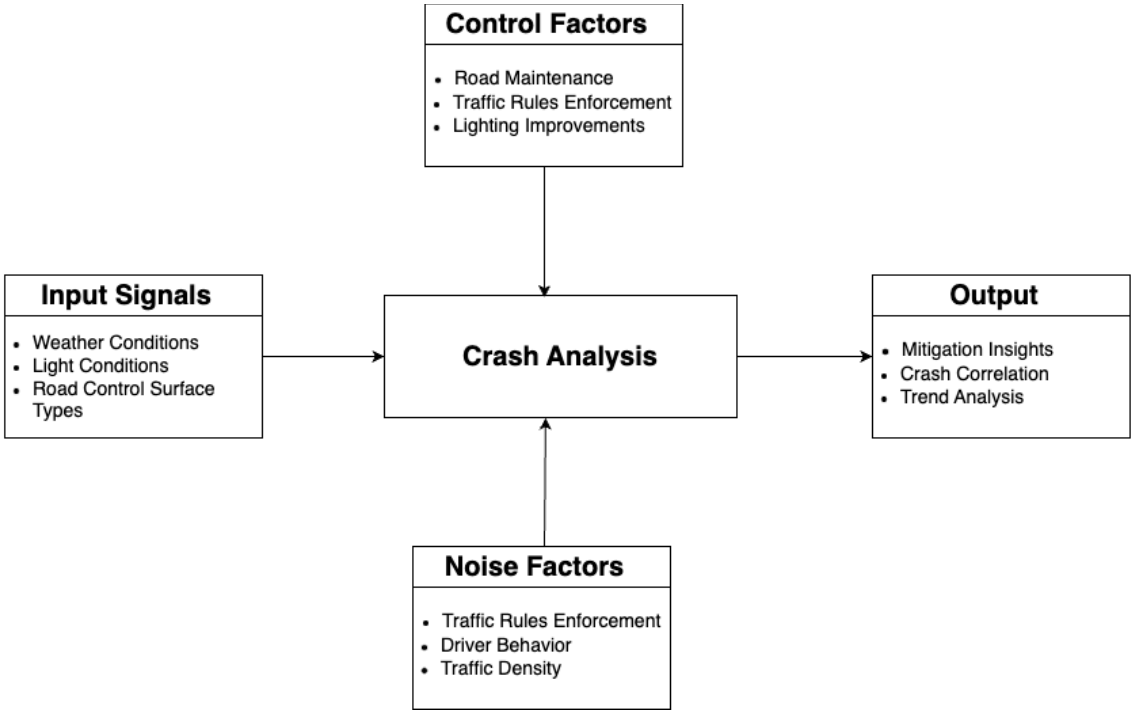
The results were carefully interpreted, with significant findings discussed in the context of road safety. Recommendations for policymakers, transportation authorities, and safety advocates were developed based on the insights derived from the analysis.

6. Scope for Future Research

While this study focuses on the environmental factors of weather, light, and road surface conditions, future work could incorporate additional dimensions such as traffic density, vehicle types, and driver behaviour. Advanced predictive modelling techniques may further enhance the understanding of crash dynamics.

This systematic methodology ensures a rigorous and transparent approach to uncovering meaningful insights from the data, contributing to evidence-based strategies for improving road safety.

3. P-Diagram


Control Factors
Road Maintenance
Traffic Rules Enforcement
Lighting Improvements
Input Signals
Weather Conditions
Light Conditions
Road Control Surface Types
Crash Analysis
Output
Mitigation Insights
Crash Correlation
Trend Analysis
Noise Factors
Traffic Rules Enforcement
Driver Behavior
Traffic Density

4. Data

The dataset used in this analysis was sourced from real-world traffic crash records and meticulously filtered to ensure accuracy and relevance. Key variables include weather conditions (e.g., Clear, Rain, Snow), road surface types (e.g., Dry, Wet, Icy), light conditions (e.g., Daylight, Night), and the number of suspected injuries. Derived variables, such as combinations of adverse weather and road surfaces (e.g., Adverse × Ideal, Non-Adverse × Non-Ideal), were added to capture interaction effects. Missing values were addressed, and irrelevant entries were excluded to maintain dataset integrity. The final dataset spans multiple years, offering a comprehensive view of how environmental factors influence injury counts, making it suitable for robust statistical and inferential analysis.

Crash Num	Date and Time	Police Sh	Crash Loc	LIGHT	Weather	Road Surf	Road Con	Traffic Co	Roadway	Trafficway	Manner of	First Harm	First Harm	Speed Lin	Category	Year	Suspected Injury	Adverse × Ide	Adverse × No	Non-Adverse	Non-Adverse × Non-Ideal
18025234	29/04/18 15:48	Day Shift (8A	FELLSWAY &	DAYLIGHT	CLOUDY	WET	UNKNOWN	NO CONTROL	NOT AT INTER	TWO-WAY, D	REAR-END	COLLISION V	ROADWAY	30	Noon	2018	6	2	4	1	2
18026134	03/05/18 11:42	Day Shift (8A	33 LINDEN A	DAYLIGHT	CLOUDY	DRY	NONE	NO CONTROL	DRIVEWAY	TWO-WAY, N	SINGLE VEHI	OTHER	OUTSIDE RO	25	Morning	2018	2	3	3	2	3
18026355	04/05/18 10:45	Day Shift (8A	243 SUMMEI	DAYLIGHT	CLOUDY	DRY	NONE	NO CONTROL	NOT AT INTER	TWO-WAY, N	SIDESWIPE,	COLLISION V	ROADWAY	30	Morning	2018	2	2	3	1	9
18028136	12/05/18 14:35	Day Shift (8A	BROADWAY	DAYLIGHT	CLOUDY	WET	NONE	TRAFFIC CON	TRAFFIC CIRC	ONE-WAY, N	ANGLE	COLLISION V	ROADWAY	30	Noon	2018	3	3	2	3	2
18028360	13/05/18 21:52	First Half (4P	BONNER AV	NIGHT	CLOUDY	DRY	NONE	STOP SIGNS	FOUR WAY I	TWO-WAY, N	ANGLE	COLLISION V	ROADWAY	30	Night	2018	3	4	3	3	7
18032697	01/06/18 13:16	Day Shift (8A	BOSTON AV	DAYLIGHT	CLOUDY	DRY	NONE	STOP SIGNS	T-INTERSECTI	TWO-WAY, N	ANGLE	COLLISION V	ROADWAY	25	Noon	2018	3		2	4	3
18033381	04/06/18 22:16	First Half (4P	BROADWAY	NIGHT	CLOUDY	WET	NONE	NO CONTROL	T-INTERSECTI	TWO-WAY, D	ANGLE	COLLISION V	ROADWAY	30	Night	2018	2		6	4	3
18033457	05/06/18 8:51	Day Shift (8A	DANE ST & VI	DAYLIGHT	CLOUDY	DRY	NONE	NO CONTROL	T-INTERSECTI	TWO-WAY, N	SIDESWIPE,	COLLISION V	ROADWAY	25	Morning	2018	1		3	3	2
18033671	06/06/18 8:03	Day Shift (8A	483 BROADW	DAYLIGHT	CLOUDY	DRY	NONE	TRAFFIC CON	NOT AT INTER	TWO-WAY, N	REAR-END	COLLISION V	ROADWAY	30	Morning	2018	4		3	2	2
18033859	07/06/18 6:43	Last Half (Mi	166 ELM STR	DAYLIGHT	CLOUDY	DRY	NONE	NO CONTROL	NOT AT INTER	TWO-WAY, N	SIDESWIPE,	COLLISION V	ROADWAY	30	Morning	2018	4		1	3	2
18035480	14/06/18 15:25	Day Shift (8A	CHERRY ST &	DAYLIGHT	CLOUDY	DRY	UNKNOWN	STOP SIGNS	FOUR WAY I	TWO-WAY, N	ANGLE	COLLISION V	ROADWAY	25	Noon	2018	2		4	3	4
18037416	23/06/18 9:01	Day Shift (8A	82 BEACON S	DAYLIGHT	CLOUDY	WET	NONE	NO CONTROL	NOT AT INTER	TWO-WAY, N	REAR-END	COLLISION V	ROADWAY	25	Morning	2018	2		2	6	2
18038577	28/06/18 18:01	First Half (4P	3 MAXWELL S	DAYLIGHT	CLOUDY	WET	UNKNOWN	NO CONTROL	NOT AT INTER	TWO-WAY, N	UNKNOWN	COLLISION V	ROADWAY	5	Evening	2018	3		2	5	2
18043554	22/07/18 16:19	First Half (4P	294 BROADW	DAYLIGHT	CLOUDY	WET	NONE	NO CONTROL	NOT AT INTER	TWO-WAY, D	REAR-END	COLLISION V	ROADWAY	30	Noon	2018	2		1	4	2
18043752	23/07/18 12:03	Day Shift (8A	244 ELM ST S	DAYLIGHT	CLOUDY	WET	NONE	NO CONTROL	NOT AT INTER	ONE-WAY, N	REAR-END	COLLISION V	ROADWAY	30	Noon	2018	4		2	2	3
18044327	26/07/18 10:11	Day Shift (8A	334 HIGHLAI	DAYLIGHT	CLOUDY	DRY	NONE	STOP SIGNS	T-INTERSECTI	TWO-WAY, N	ANGLE	COLLISION V	ROADWAY	25	Morning	2018	5		6	7	4
18047038	08/08/18 13:28	Day Shift (8A	582 BROADW	DAYLIGHT	CLOUDY	DRY	NONE	TRAFFIC CON	NOT AT INTER	TWO-WAY, N	SINGLE VEHI	OTHER	OUTSIDE RO	30	Noon	2018	3		6	3	3
18047967	12/08/18 20:09	First Half (4P	HEATH & TEM	NIGHT	CLOUDY	DRY	NONE	STOP SIGNS	FOUR WAY I	TWO-WAY, N	ANGLE	COLLISION V	ROADWAY	25	Evening	2018	4		4	3	3
18050269	23/08/18 11:42	Day Shift (8A	BROADWAY	DAYLIGHT	CLOUDY	DRY	NONE	TRAFFIC CON	FOUR WAY I	TWO-WAY, D	REAR-END	COLLISION V	ROADWAY	30	Morning	2018	2		5	4	5
18056033	19/09/18 14:09	Day Shift (8A	BROADWAY	DAYLIGHT	CLOUDY	DRY	NONE	TRAFFIC CON	T-INTERSECTI	TWO-WAY, N	ANGLE	COLLISION V	ROADWAY	30	Noon	2018	3		2	4	2
18056080	19/09/18 17:05	First Half (4P	LOWELL ST &	DAYLIGHT	CLOUDY	DRY	NONE	NO CONTROL	T-INTERSECTI	TWO-WAY, N	SINGLE VEHI	COLLISION V	OUTSIDE RO	20	Evening	2018	2		3	3	3
18058644	01/10/18 16:40	First Half (4P	ALBION ST &	DAYLIGHT	CLOUDY	WET	NONE	STOP SIGNS	FOUR WAY I	TWO-WAY, N	ANGLE	COLLISION V	ROADWAY	20	Noon	2018	3		4	3	
18058977	03/10/18 8:39	Day Shift (8A	21 UNION SC	DAYLIGHT	CLOUDY	DRY	TRAFFIC CON	TRAFFIC CON	FOUR WAY I	TWO-WAY, D	REAR-END	COLLISION V	ROADWAY	20	Morning	2018	3		3	2	
18059739	07/10/18 10:37	Day Shift (8A	SHORE DR &	DAYLIGHT	CLOUDY	DRY	NONE	STOP SIGNS	T-INTERSECTI	TWO-WAY, N	REAR-END	COLLISION V	ROADSIDE	30	Morning	2018	5		2	2	
18060039	08/10/18 21:34	First Half (4P	11 COLLEGE	NIGHT	CLOUDY	WET	NONE	NO CONTROL	NOT AT INTER	TWO-WAY, N	SIDESWIPE,	COLLISION V	ROADSIDE	30	Night	2018	2		3	4	
18060093	09/10/18 6:43	Last Half (Mi	BROADWAY	DAYLIGHT	CLOUDY	DRY	NONE	TRAFFIC CON	FOUR WAY I	TWO-WAY, N	ANGLE	COLLISION V	ROADWAY	30	Morning	2018	1		3	3	
18060712	12/10/18 7:00	Last Half (Mi	16 BEACON S	DAWN	CLOUDY	WET	NONE	NO CONTROL	T-INTERSECTI	TWO-WAY, N	ANGLE	COLLISION V	ROADWAY	30	Morning	2018	3		3	3	
18060921	13/10/18 6:52	Last Half (Mi	BROADWAY	DAWN	CLOUDY	DRY	UNKNOWN	STOP SIGNS	T-INTERSECTI	TWO-WAY, N	ANGLE	COLLISION V	ROADWAY	30	Morning	2018	3		4	2	
18060987	13/10/18 13:15	Day Shift (8A	493 SOMERV	DAYLIGHT	CLOUDY	WET	NONE	STOP SIGNS	DRIVEWAY	TWO-WAY, N	ANGLE	COLLISION V	ROADWAY	20	Noon	2018	3		2	1	
18061372	15/10/18 8:30	Day Shift (8A	363 REVOLU	DAYLIGHT	CLOUDY	DRY	NONE	NO CONTROL	NOT AT INTER	TWO-WAY, N	ANGLE	COLLISION V	ROADWAY	25	Morning	2018	3		3	3	
18063104	24/10/18 16:32	First Half (4P	SOMERVILL E	DAYLIGHT	CLOUDY	WET	NONE	NO CONTROL	NOT AT INTER	UNKNOWN	ANGLE	COLLISION V	ROADWAY	25	Noon	2018	9		4	3	
18065958	06/11/18 12:45	Day Shift (8A	259 SOMERV	DAYLIGHT	CLOUDY	WET	NONE	NO CONTROL	T-INTERSECTI	TWO-WAY, N	UNKNOWN	COLLISION V	ROADWAY	20	Noon	2018	2		2	4	
18066654	09/11/18 14:47	Day Shift (8A	51 WARREN S	DAYLIGHT	CLOUDY	DRY	NONE	NO CONTROL	NOT AT INTER	ONE-WAY, N	UNKNOWN	COLLISION V	ROADWAY	25	Noon	2018	2		2	3	
18069042	21/11/18 13:02	Day Shift (8A	HOLLAND ST	DAYLIGHT	CLOUDY	DRY	NONE	STOP SIGNS	T-INTERSECTI	TWO-WAY, N	ANGLE	COLLISION V	ROADWAY	20	Noon	2018	3		3	2	
18069838	26/11/18 8:26	Day Shift (8A	NORTH ST & F	DAYLIGHT	CLOUDY	WET	TRAFFIC CON	TRAFFIC CON	FOUR WAY I	TWO-WAY, N	SIDESWIPE,	COLLISION V	ROADSIDE	20	Morning	2018	1		3	3	
18069921	26/11/18 15:01	Day Shift (8A	PRESCOTT ST	DAYLIGHT	CLOUDY	WET	NONE	STOP SIGNS	FOUR WAY I	TWO-WAY, D	ANGLE	COLLISION V	ROADWAY	20	Noon	2018	3		4	3	
18070671	29/11/18 21:46	First Half (4P	220 WASHIN	NIGHT	CLOUDY	DRY	NONE	NO CONTROL	T-INTERSECTI	TWO-WAY, D	ANGLE	COLLISION V	ROADWAY	25	Night	2018	3			5	
18071302	03/12/18 7:40	Last Half (Mi	FITCHBURG S	DAYLIGHT	CLOUDY	WET	NONE	NO CONTROL	T-INTERSECTI	TWO-WAY, N	SIDESWIPE,	COLLISION V	ROADWAY	25	Morning	2018	5			2	
18073581	14/12/18 15:58	Day Shift (8A	532 MEDFOF	DAYLIGHT	CLOUDY	DRY	NONE	NO CONTROL	NOT AT INTER	TWO-WAY, N	SINGLE VEHI	COLLISION V	OUTSIDE RO	15	Noon	2018	2			3	
18075098	22/12/18 0:31	Last Half (Mi	259 MCGRAT	NIGHT	CLOUDY	WET	NONE	NO CONTROL	NOT AT INTER	ONE-WAY, N	SINGLE VEHI	COLLISION V	SHOULDER-	35	Night	2018	2			2	
19000406	03/01/19 6:36	Last Half (Mi	DANE ST & SC	NIGHT	CLOUDY	WET	NONE	TRAFFIC CON	FOUR WAY I	TWO-WAY, N	ANGLE	COLLISION V	ROADWAY	30	Morning	2019	7			4	
19001660	10/01/19 9:19	Day Shift (8A	367 MEDFOF	DAYLIGHT	CLOUDY	DRY	NONE	NO CONTROL	NOT AT INTER	TWO-WAY, N	SINGLE VEHI	COLLISION V	OUTSIDE RO	25	Morning	2019	2			5	
19002784	16/01/19 12:58	Day Shift (8A	GLENWOOD	DAYLIGHT	CLOUDY	DRY	NONE	STOP SIGNS	FOUR WAY I	TWO-WAY, N	ANGLE	COLLISION V	ROADWAY	25	Noon	2019	2			3	
19004102	23/01/19 10:29	Day Shift (8A	BROADWAY	DAYLIGHT	CLOUDY	WET	UNKNOWN	TRAFFIC CON	FOUR WAY I	TWO-WAY, D	UNKNOWN	COLLISION V	ROADWAY	25	Morning	2019	5			1	
19006234	04/02/19 0:11	Last Half (Mi	26 UNION SC	NIGHT	CLOUDY	WET	NONE	NO CONTROL	NOT AT INTER	TWO-WAY, D	SIDESWIPE,	COLLISION V	ROADWAY	20	Night	2019	1			1	
19006750	06/02/19 16:04	First Half (4P	GROVE ST &	DAYLIGHT	CLOUDY	DRY	NONE	NO CONTROL	FOUR WAY I	TWO-WAY, N	ANGLE	COLLISION V	ROADWAY	30	Noon	2019	4			2	
19007054	08/02/19 7:48	Last Half (Mi	BERKELEY ST	DAYLIGHT	CLOUDY	WET	NONE	NO CONTROL	T-INTERSECTI	ONE-WAY, N	ANGLE	COLLISION V	ROADWAY	20	Morning	2019	6			3	
19007825	12/02/19 11:48	Day Shift (8A	1 FRANEY RD	DAYLIGHT	CLOUDY	DRY	NONE	NO CONTROL	NOT AT INTER	TWO-WAY, N	ANGLE	COLLISION V	OUTSIDE RO	30	Morning	2019	4			3	
19007861	12/02/19 15:11	Day Shift (8A	GLENDALE A	DAYLIGHT	CLOUDY	SNOW	ROAD SURFA	STOP SIGNS	T-INTERSECTI	TWO-WAY, N	ANGLE	COLLISION V	ROADWAY	30	Noon	2019	3			3	
19008039	13/02/19 15:24	Day Shift (8A	8 MELVIN ST	DAYLIGHT	CLOUDY	WET	NONE	NO CONTROL	NOT AT INTER	ONE-WAY, N	SINGLE VEHI	COLLISION V	OUTSIDE RO	25	Noon	2019	3			1	
19012188	05/03/19 17:44	First Half (4P	166 ELM ST S	NIGHT	CLOUDY	WET	NONE	NO CONTROL	NOT AT INTER	TWO-WAY, D	SIDESWIPE,	COLLISION V	ROADWAY	20	Evening	2019	2			2	
19014338	15/03/19 13:32	Day Shift (8A	445 REVOLU	DAYLIGHT	CLOUDY	DRY	NONE	NO CONTROL	NOT AT INTER	TWO-WAY, N	SIDESWIPE,	COLLISION V	ROADSIDE	25	Noon	2019	2			5	
19014426	15/03/19 19:21	First Half (4P	BELMONT ST	NIGHT	CLOUDY	DRY	NONE	STOP SIGNS	T-INTERSECTI	TWO-WAY, N	ANGLE	COLLISION V	ROADWAY	30	Evening	2019	3			3	
19015865	22/03/19 14:50	Day Shift (8A	62 DANE ST S	DAYLIGHT	CLOUDY	WET	NONE	NO CONTROL	NOT AT INTER	TWO-WAY, N	REAR-END	COLLISION V	ROADWAY	25	Noon	2019	4			2	
19016305	25/03/19 7:37	Last Half (Mi	360 FOLEY S	DAYLIGHT	CLOUDY	DRY	NONE	NO CONTROL	NOT AT INTER	TWO-WAY, N	ANGLE	COLLISION V	ROADWAY	25	Morning	2019	5			2	
19018697	05/04/19 17:27	First Half (4P	HUDSON ST	DAYLIGHT	CLOUDY	DRY	NONE	STOP SIGNS	FOUR WAY I	TWO-WAY, N	ANGLE	COLLISION V	ROADWAY	25	Evening	2019	2			5	
19019189	08/04/19 8:33	Day Shift (8A	464 BROADW	DAYLIGHT	CLOUDY	WET	NONE	NO CONTROL	NOT AT INTER	TWO-WAY, N	ANGLE	COLLISION V	ROADWAY	25	Morning	2019	4			2	
19019253	08/04/19 15:40	Day Shift (8A	CENTRAL ST	DAYLIGHT	CLOUDY	WET	NONE	NO CONTROL	FOUR WAY I	TWO-WAY, N	REAR-END	COLLISION V	ROADWAY	25	Noon	2019	5			4	
19019432	09/04/19 11:23	Day Shift (8A	51 WARREN S	DAYLIGHT	CLOUDY	DRY	NONE	NO CONTROL	NOT AT INTER	ONE-WAY, N	SIDESWIPE,	COLLISION V	ROADWAY	25	Morning	2019	2			2	
19021143	18/04/19 7:26	Last Half (Mi	WEBSTER AV	DAYLIGHT	CLOUDY	DRY	NONE	STOP SIGNS	FOUR WAY I	TWO-WAY, N	ANGLE	COLLISION V	ROADWAY	25	Morning	2019	3			4	
19021199	18/04/19 11:06	Day Shift (8A	518 MEDFOF	DAYLIGHT	CLOUDY	DRY	NONE	NO CONTROL	NOT AT INTER	TWO-WAY, N	SINGLE VEHI	COLLISION V	ROADWAY	25	Morning	2019	4			5	
19022392	24/04/19 8:49	Day Shift (8A	486 SOMERV	DAYLIGHT	CLOUDY	DRY	NONE	NO CONTROL	NOT AT INTER	TWO-WAY, D	REAR-END	COLLISION V	ROADWAY	30	Morning	2019	3			2	
19022921	26/04/19 13:22	Day Shift (8A	CENTRAL ST	DAYLIGHT	CLOUDY	WET	NONE	NO CONTROL	T-INTERSECTI	TWO-WAY, N	ANGLE	COLLISION V	ROADWAY	20	Noon	2019	4			1	
19023565	29/04/19 18:57	First Half (4P	ELM ST & POI	DAYLIGHT	CLOUDY	DRY	NONE	NO CONTROL	T-INTERSECTI	TWO-WAY, N	ANGLE	COLLISION V	ROADWAY	30	Evening	2019	3			2	
19023984	01/05/19 17:34	First Half (4P	136 PACKARD	DAYLIGHT	CLOUDY	DRY	NONE	NO CONTROL	DRIVEWAY	TWO-WAY, N	ANGLE	COLLISION V	ROADWAY	25	Evening	2019	2			1	

5. Descriptive Statistics: Distributions

Effect of light conditions on the number of injuries:

The following analysis shows the effect of light conditions (Daylight, Night, Dusk and Dawn) on the number of injuries. For this analysis, we have taken a dataset of all the crashes that have occurred in Somerville between 2018 and 2024. The following data gives us details like mean, standard deviation, and range of the number of accidents throughout the given time period. A few interesting things about the effect of light conditions on the occurrence of accidents and number of injuries are given as follows:

- Surprisingly, the percentage of accidents that have occurred during daylight is the highest with about 4341 injuries and 1559 accidents.
- As compared to most light conditions, it is seen that the least number of accidents and injuries have taken place during dawn with 57 injuries and 22 accidents in total throughout the course of 6 years.
- The crash analysis shows that about 169 injuries have happened in about 57 accidents that took place during dusk.
- About 1858 injuries and 641 accidents have been recorded during night as seen from the data available.

Light Conditions	Number of suspected injuries	Percentage of injuries	Number of accidents	Percentage of accidents
DAYLIGHT	4341	67.5326696	1559	68.4071961
DAWN	57	0.88674549	22	0.96533567
DUSK	169	2.62912259	57	2.50109697
NIGHT	1858	28.9047915	641	28.1263712

Analysis	
Mean	2.820535
Standard Error	0.024303
Median	3
Mode	3
Standard Deviation	1.160439
Sample Variance	1.346028
Kurtosis	1.93822
Skewness	0.938456
Range	9
Minimum	0
Maximum	9
Sum	6428
Count	2279

The crash data gives details of an average of 2.82 injuries every year with a maximum of 9 injuries and deviation of 1.16.

Effect of weather conditions on the number of injuries:

The following analysis shows the effect of weather conditions (Clear, Cloudy, Rain and Snow) on the number of injuries. For this analysis, we have considered a dataset of all the crashes that have occurred in Somerville from 2018 to 2024. The following data analysis presents details like mean, standard deviation, and range of the number of accidents throughout the given time period of 6 years. A few interesting things about how the weather conditions affect the occurrence and possibility of accidents and injuries are given as follows:

- The analysis shows an astonishingly high percentage of accidents that have occurred during clear weather conditions with about 4640 injuries and 1655 accidents.
- As compared to most weather conditions, the least number of accidents and injuries have taken place during snow with 157 injuries and 54 accidents in total between 2018 and 2024.
- There have been about 962 injuries in the 331 accidents that have been recorded in cloudy weather suggesting pretty adverse conditions that affect the number and kind of accidents.
- The crash analysis shows that rain has been a major factor in the crashes amounting to about 669 injuries in 239 accidents that have taken place.

Weather Conditions	Number of suspected injuries	Percentage of injuries	Number of accidents	Percentage of accidents
CLEAR	4640	72.1841942	1655	72.61957
CLOUDY	962	14.9657747	331	14.523914
RAIN	669	10.4075918	239	10.4870557
SNOW	157	2.44243933	54	2.36946029

Analysis	
Mean	2.820535
Standard Error	0.024303
Median	3
Mode	3
Standard Deviation	1.160439
Sample Variance	1.346028
Kurtosis	1.93822
Skewness	0.938456
Range	9
Minimum	0
Maximum	9
Sum	6428
Count	2279

The crash data gives details of an average of 2.82 injuries every year with a maximum of 9 injuries and deviation of 1.16.

Effect of road surface on the number of injuries:

The following analysis shows the effect of road surface (Dry, Ice, Sand, Mud, Dirt, Oil, Gravel, Snow, Wet) on the number of injuries that occurred between 2018 and 2024. For this analysis, we have considered a dataset of all the crashes that have occurred in Somerville over a course of 6 years. The following data analysis presents details like mean, standard deviation, and range of the number of accidents throughout the given time period. A few interesting things about the effect of road surface on the occurrence and possibility of accidents and injuries are given as follows:

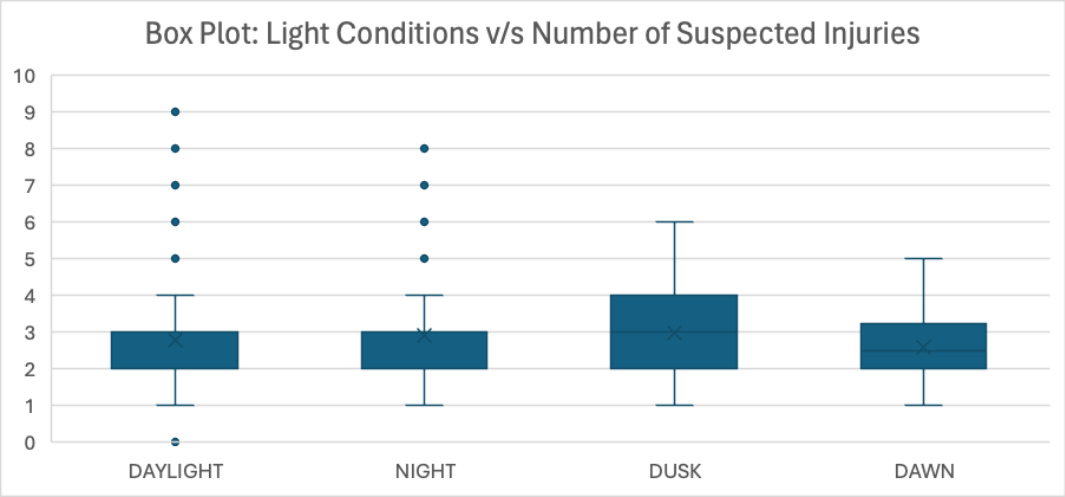
- As absurd it might sound, it can be derived from the analysis of the dataset, that dry road surfaces have resulted in a majority of the accidents throughout the 6 year time period with 5118 injuries in the 1828 accidents that have occurred throughout.
- Its surpising how factors like sand, mud, dirt, oil and gravel have only affected about 0.187% of the injuries and 0.175% of the accidents between 2018 and 2024.
- There have been about 41 injuries in the 16 accidents that have been recorded on icy road surfaces which suggests that a rather small percentage of accidents have been caused due to formation of ice on roads.
- Snow has been a contributing factor for a few of the accidents with 151 injuries recorded throughout the time period.
- The crash analysis depicts the possibility of wet roads being a major factor in contributing to road accidents with about 1106 injuries and 381 accidents

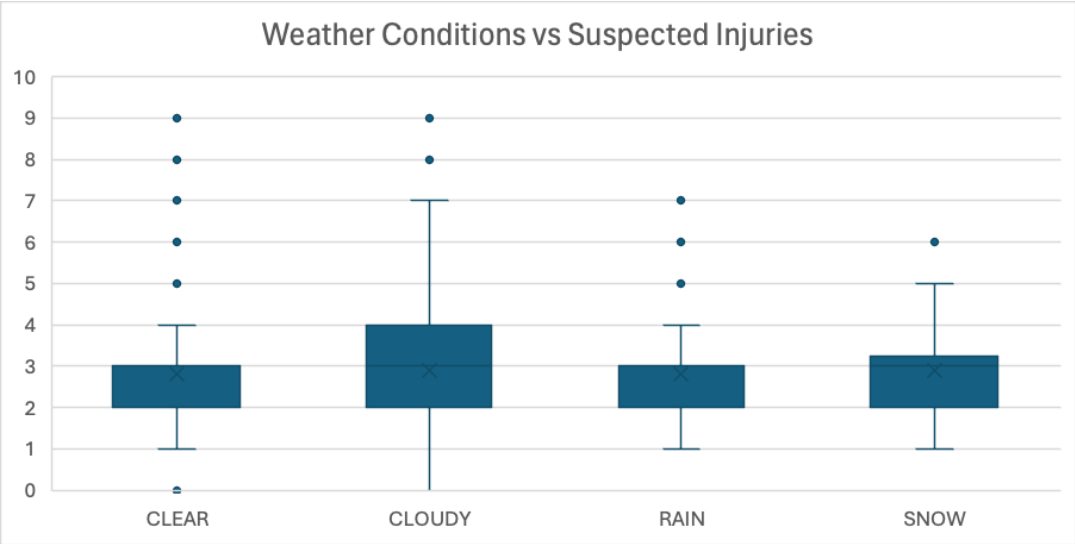
Road Surface	Number of suspected injuries	Percentage of injuries	Number of accidents	Percentage of accidents
DRY	5118	79.6204107	1828	80.2106187
ICE	41	0.63783447	16	0.70206231
SAND, MUD, DIRT, OIL, GRAVEL	12	0.18668326	4	0.17551558
SNOW	151	2.3490977	50	2.19394471
WET	1106	17.2059739	381	16.7178587

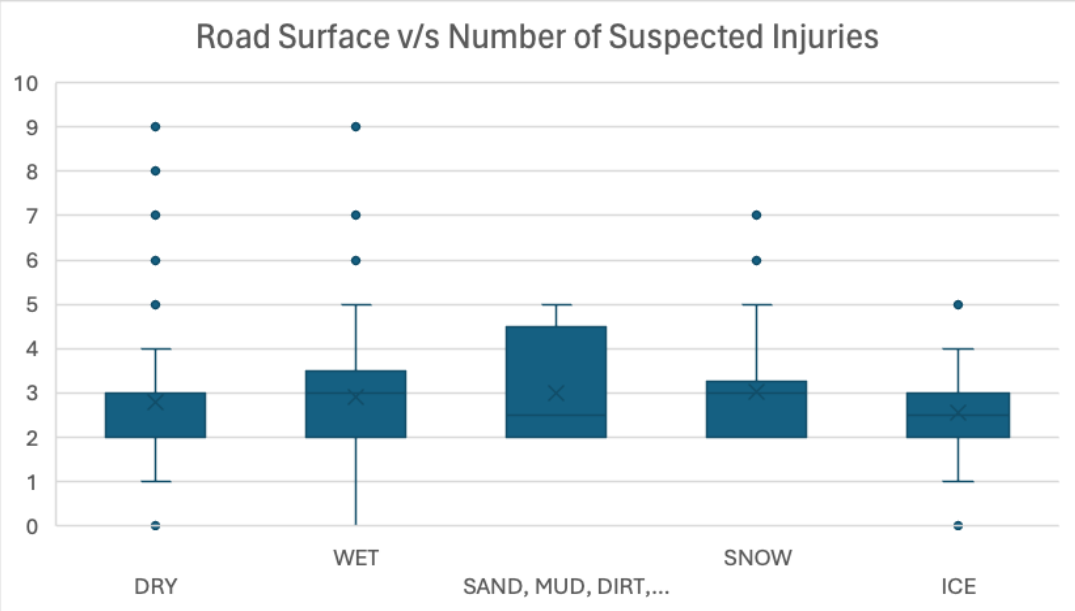
Analysis	
Mean	2.820535
Standard Error	0.024303
Median	3
Mode	3
Standard Deviation	1.160439
Sample Variance	1.346028
Kurtosis	1.93822
Skewness	0.938456
Range	9
Minimum	0
Maximum	9
Sum	6428
Count	2279

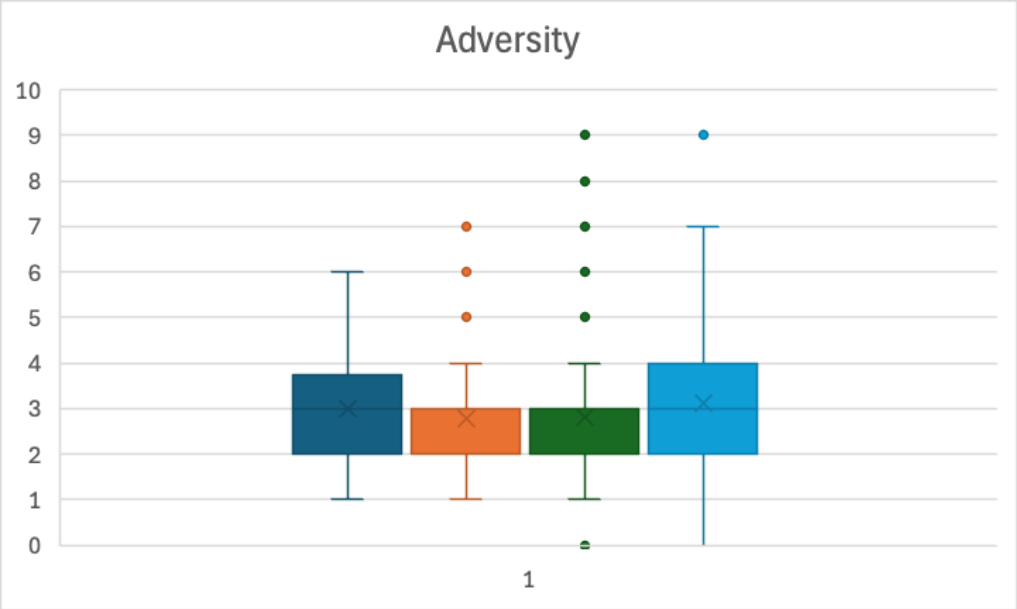
The crash data gives details of an average of 2.82 injuries every year with a maximum of 9 injuries and deviation of 1.16.

6. Descriptive Statistics: Box Plot

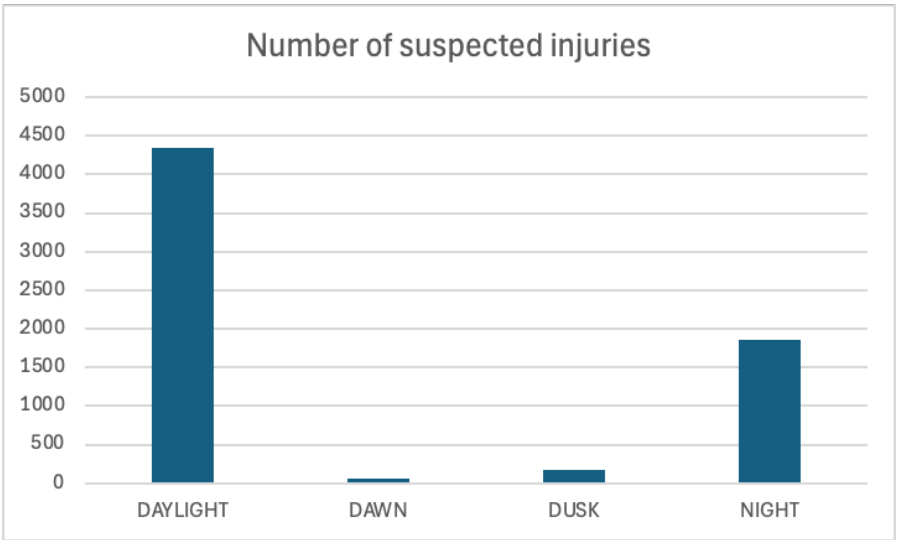


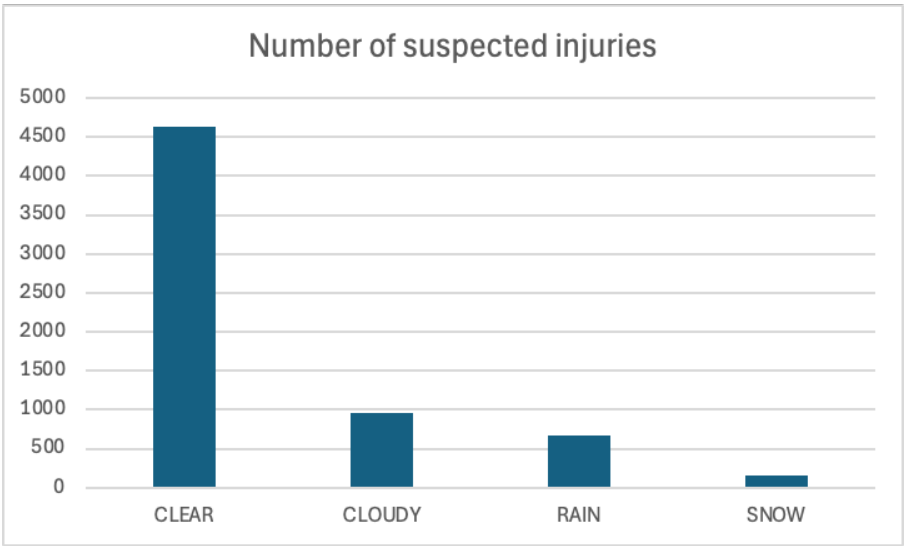


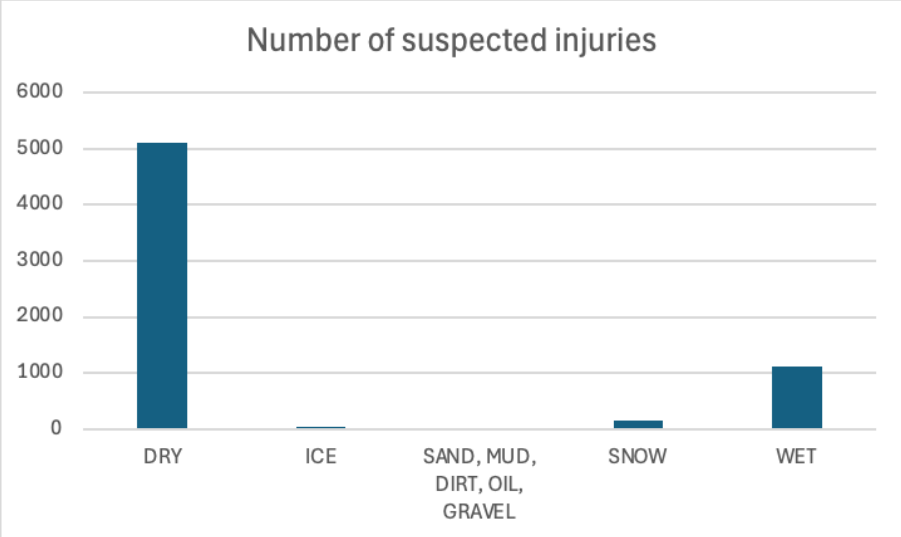


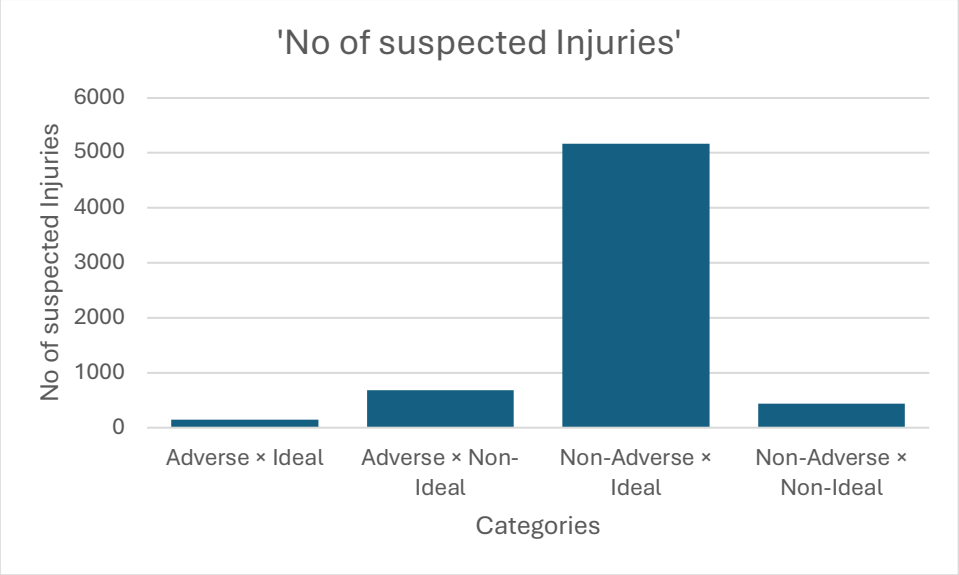

Adversity
10
9
8
7
6
5
4
3
2
1
0
1

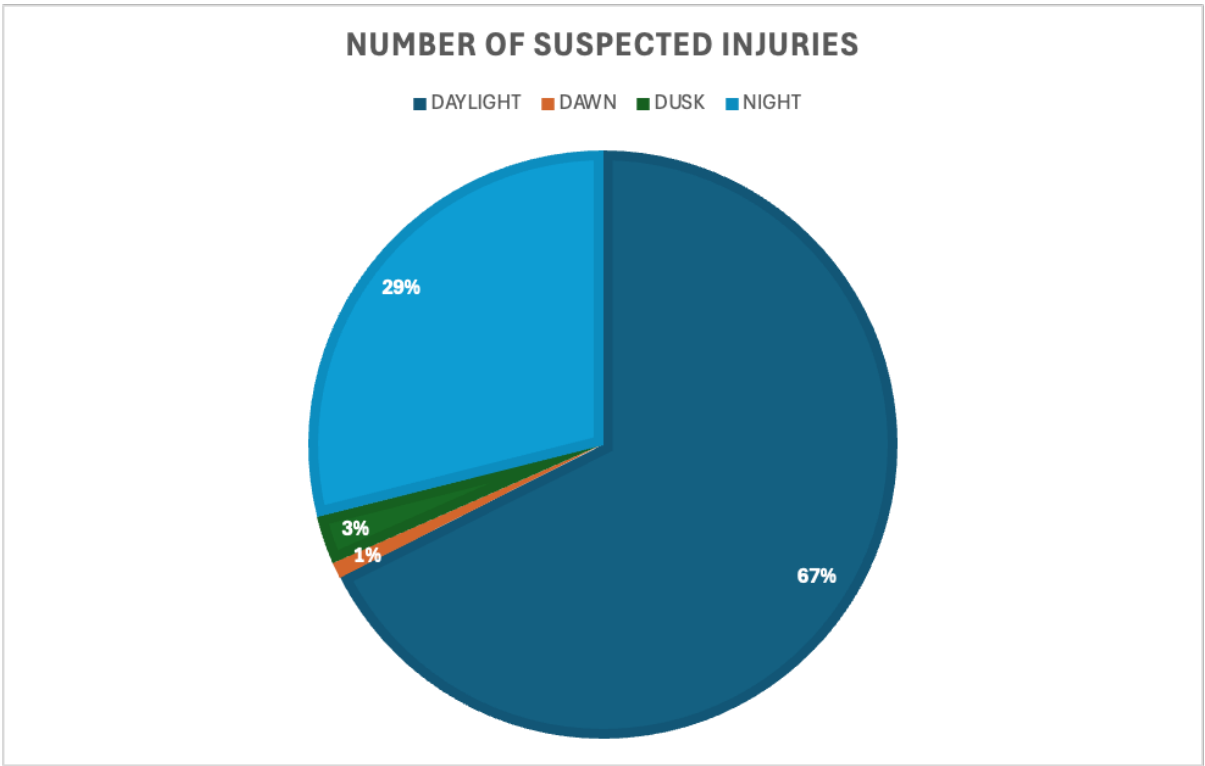
7. Descriptive Statistics: Graph





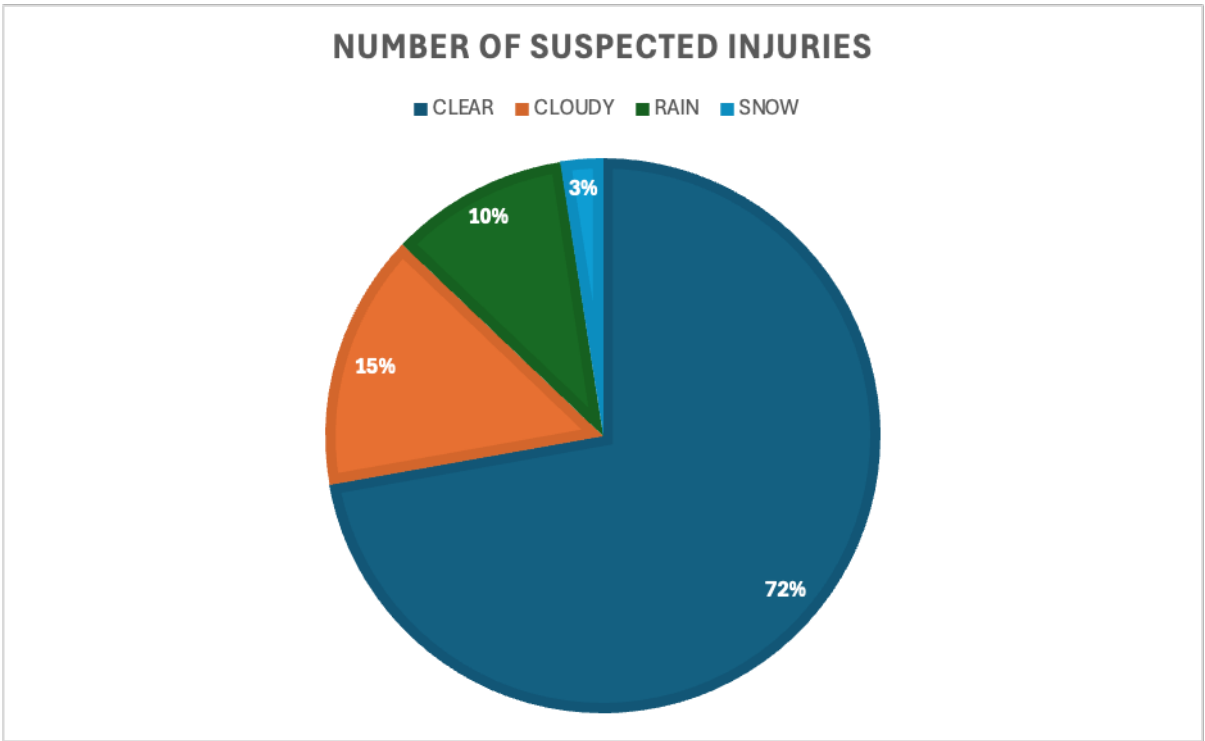



'No of suspected Injuries'
No of suspected Injuries
6000
5000
4000
3000
2000
1000
0
Adverse × Ideal
Adverse × Non-Ideal
Non-Adverse × Ideal
Non-Adverse × Non-Ideal
Categories



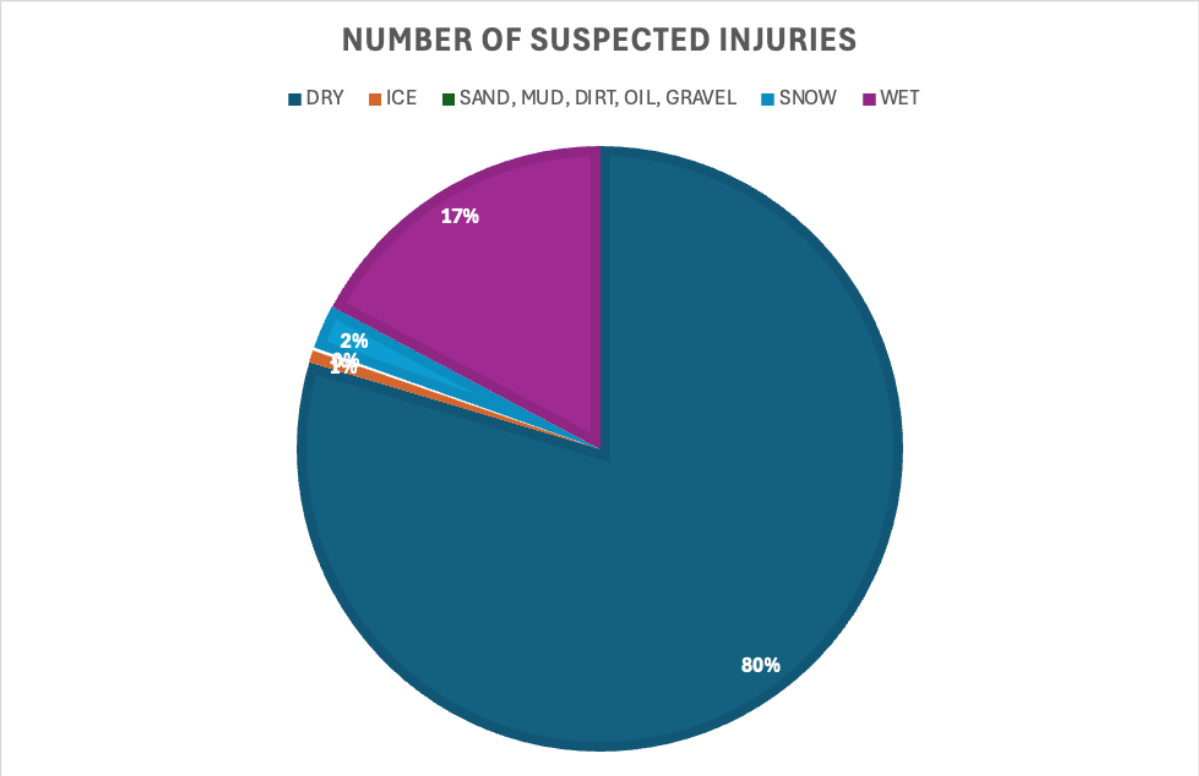
Time of Day Distribution

The chart shows that most suspected injuries (67%) occur during daylight hours, followed by nighttime incidents at 29%. Dawn and dusk periods account for much smaller percentages at 1% and 3% respectively.



Weather Conditions

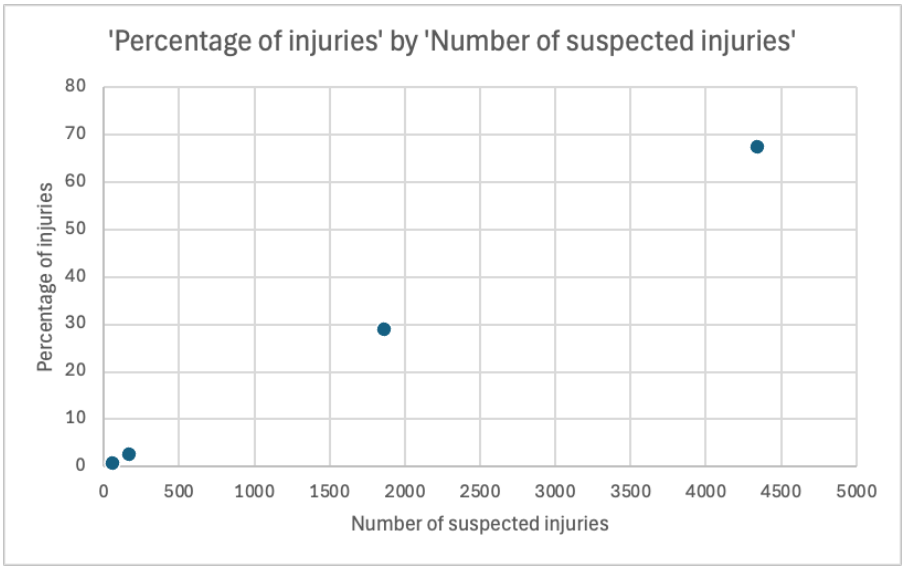
The chart reveals that clear weather dominates injury incidents at 72% of cases. Cloudy conditions account for 15%, while rain represents 10% of cases. Snow-related injuries make up only 3% of the total.



Road Surface Conditions

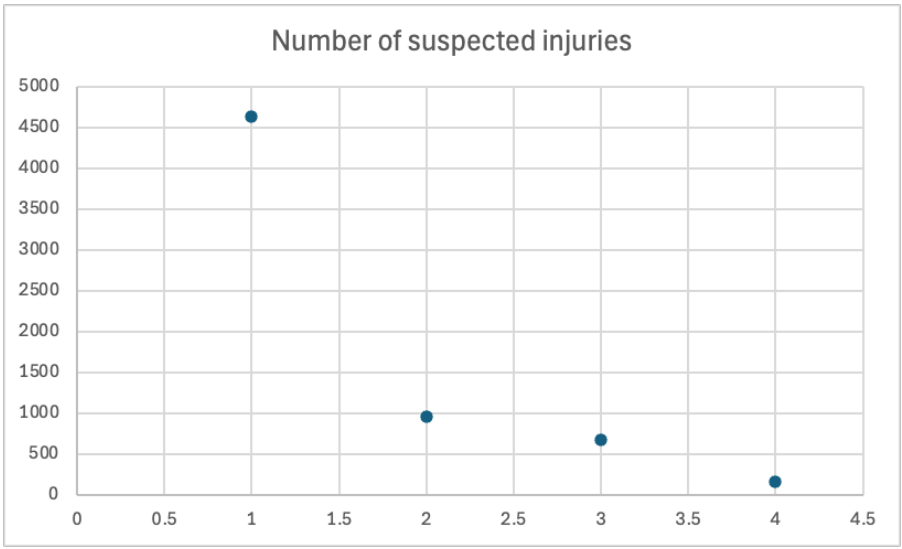
The chart breaks down injuries by road surface conditions, with dry conditions being predominant at 80%. Wet surfaces account for 17% of cases, while snow, ice, and various road debris (sand, mud, dirt, oil, gravel) each represent 2% or less of total incidents.

The data from the pie graphs suggest that most injuries occur during daylight hours, in clear weather, and on dry road conditions, rather than during adverse conditions as might be expected.



Light Conditions vs Suspected Injuries

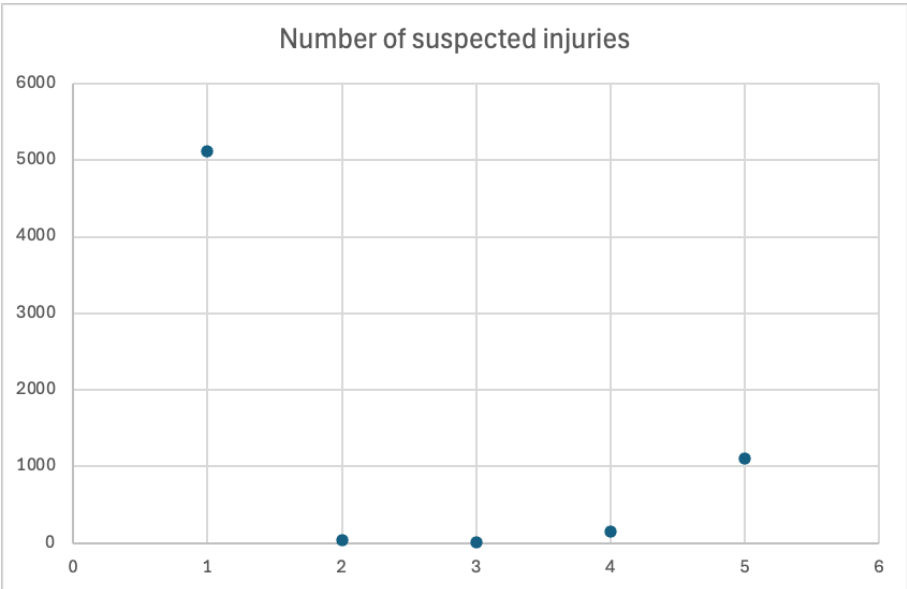
The data shows 4,500 injuries in bright light (condition 1), decreasing to 1,000 in moderate light (condition 2), 650 in dim light (condition 3), and 200 in dark conditions (condition 4). Most injuries (67%) occur in bright lighting conditions.



Weather Conditions vs Suspected Injuries

The data shows a clear negative correlation between weather conditions and suspected injuries:

Clear weather shows 4,500 injuries, followed by rain (1,000), snow (650), and fog (200). Injury rates decrease as weather conditions worsen, likely due to increased caution.



Road Surface vs Suspected injuries

The graph shows the distribution of suspected injuries across 5 road surface conditions. The highest number of injuries (approximately 5,000) occurs at condition 1, representing dry/normal roads. Conditions 2-4 show minimal injuries (near zero),

while condition 5 shows about 1,000 injuries. This suggests that most incidents occur during normal road conditions.

8. Statistical Analysis

Analysis of Variance (ANOVA) is a statistical method used to compare the means of three or more groups to determine if there are statistically significant differences among them. By analysing the variation within and between groups, ANOVA helps identify whether observed differences in sample means are due to chance or a systematic factor. In this study, one-way ANOVA was applied to assess the impact of weather conditions on the number of suspected injuries in traffic crashes.

Hypothesis:

- **Null Hypothesis (H_0):** The mean number of suspected injuries does not differ across weather conditions.
- **Alternative Hypothesis (H_1):** The mean number of suspected injuries differs across weather conditions.

Results:

Anova: Single Factor

SUMMARY

Groups	*Count*	*Sum*	*Average*	*Variance*
CLEAR	1655	4640	2.80362538	1.33323957
CLOUDY	331	962	2.90634441	1.50332326
RAIN	239	669	2.79916318	1.23680602
SNOW	54	157	2.90740741	1.29315164

ANOVA

urce of Variati	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Between Gro	3.42715276	3	1.14238425	0.8481653	0.46745725	2.60881434
Within Group	3064.17179	2275	1.3468887			
Total	3067.59895	2278				

P-value = 0.467:

- The p-value is much greater than the significance level (α=0.05\alpha = 0.05α=0.05).

- **Interpretation**: Fail to reject the null hypothesis. There is no significant difference in the mean number of suspected injuries across weather conditions.

F-statistic = 0.8481 vs. F critical = 2.6088:

- The F-statistic is less than the F critical value.
- **Interpretation**: The variation between the groups (weather conditions) is not large enough to suggest significant differences compared to the variation within groups.

Hypothesis

To evaluate whether light conditions (Daylight, Night, Dawn, Dusk) significantly affect the mean number of suspected injuries in traffic crashes, the following hypotheses were formulated:

- **Null Hypothesis (H_0):** The mean number of suspected injuries does not differ across light conditions.
- **Alternative Hypothesis (H_1):** The mean number of suspected injuries differs across light conditions.

Results:

Anova: Single Factor

SUMMARY

Groups	*Count*	*Sum*	*Average*	*Variance*
DAYLIGHT	1559	4383	2.81141758	1.33540229
NIGHT	2272	6404	2.81866197	1.34534951
DAWN	2133	6007	2.81622128	1.32643468
DUSK	2189	6166	2.81681133	1.32611579

ANOVA

urce of Variati	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Between Gro	0.05021904	3	0.01673968	0.01255475	0.99807758	2.60599878
Within Group	10865.3456	8149	1.33333484			
Total	10865.3958	8152				

P-value = 0.9981:

- The p-value is much greater than the significance level (α=0.05\alpha = 0.05α=0.05).

- **Interpretation**: Fail to reject the null hypothesis. There is no significant difference in the mean number of suspected injuries across light conditions (Daylight, Night, Dawn, Dusk).

F-statistic = 0.0126 vs. F critical = 2.6060:

- The F-statistic is far smaller than the F critical value.
- **Interpretation**: The variation between the groups (light conditions) is negligible compared to the variation within the groups.

Hypothesis

To evaluate whether road surface conditions (Dry, Ice, Sand, Snow) significantly affect the mean number of suspected injuries in traffic crashes, the following hypotheses were tested:

- **Null Hypothesis (H_0):** The mean number of suspected injuries does not differ across road surface conditions.
- **Alternative Hypothesis (H_1):** The mean number of suspected injuries differs across road surface conditions.

Results:

Anova: Single Factor

SUMMARY

Groups	*Count*	*Sum*	*Average*	*Variance*
DRY	1828	5118	2.79978118	1.31402294
ICE	1913	5328	2.78515421	1.29116024
SAND	1245	3542	2.84497992	1.39636746
SNOW	1982	5529	2.78960646	1.30805951

ANOVA

urce of Variati	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Between Gro	3.14843905	3	1.04947968	0.7946035	0.4966997	2.60618422
Within Group	9197.76531	6964	1.32075895			
Total	9200.91375	6967				

P-value = 0.4967:

- The p-value is greater than the significance level (α=0.05\alpha = 0.05α=0.05).
- **Interpretation**: Fail to reject the null hypothesis. There is no significant difference in the mean number of suspected injuries across road surface conditions (Dry, Ice, Sand, Snow).

F-statistic = 0.7946 vs. F critical = 2.6062:

- The F-statistic is smaller than the F critical value.
- **Interpretation**: The variation between the groups (road surface conditions) is not significant compared to the variation within the groups.

Our Hypothesis

"The number of injuries is significantly higher on non-ideal road surfaces during adverse weather conditions."

Anova: Single Factor

SUMMARY

Groups	*Count*	*Sum*	*Average*	*Variance*
Adverse × Ide	48	143	2.97916667	1.29742908
Adverse × No	245	683	2.7877551	1.23345601
Non-Adverse	1846	5167	2.79902492	1.32273076
Non-Adverse	140	435	3.10714286	1.80858171

ANOVA

urce of Variati	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Between Gro	13.8254129	3	4.60847098	3.4332184	0.01634	2.60881434
Within Group	3053.77353	2275	1.34231804			
Total	3067.59895	2278				

Null Hypothesis (H_0):

There is no significant difference in the number of injuries across combinations of weather conditions (Adverse, Non-Adverse) and road surface conditions (Ideal, Non-Ideal).

Alternative Hypothesis (H_1):

There is a significant difference in the number of injuries across these combinations, particularly for adverse weather combined with non-ideal surfaces.

Key Findings

ANOVA Results:

- **P-value = 0.01634 (< 0.05)**:
  - We reject the null hypothesis.
  - This indicates a statistically significant difference in the number of injuries across the groups (Adverse × Ideal, Adverse × Non-Ideal, Non-Adverse × Ideal, Non-Adverse × Non-Ideal).

Group Analysis:

- **Highest Average Injuries:**
  - The group with the highest average number of injuries is **Non-Adverse × Non-Ideal (3.107)**.
- **Interpretation:**
  - Non-ideal road surfaces (e.g., wet, icy) play a major role in increasing the number of injuries, even when the weather is not adverse.

Key Observations

- Non-ideal road surfaces, such as wet or icy roads, pose a greater risk of injuries irrespective of weather conditions.
- Adverse weather combined with ideal surfaces does not significantly increase injury rates compared to non-adverse conditions.

9. Conclusion

General Findings

1. The study concludes that while weather, light, and road surface conditions independently show no significant differences in injury counts, the combination of adverse weather and non-ideal road surfaces creates a notable increase in injury risks. This suggests that isolated environmental factors may not fully capture the complexities of traffic crash dynamics.

2. **Practical Implications**

a. **Road Maintenance**: Prioritizing improvements to road surfaces, especially in high-risk areas, could significantly reduce injuries.
b. **Public Awareness**: Targeted safety campaigns should highlight the dangers of driving on non-ideal surfaces during all weather conditions.
c. **Policy Recommendations**: Transportation authorities should consider integrating these findings into strategic road safety plans, emphasizing preventive measures during adverse weather.

10. Suggestions

1. **Data Quality Improvement**
    a. **Standardized Reporting**: Advocate for uniform crash reporting protocols across regions to ensure consistency in environmental and injury data.
    b. **Enhanced Dataset Features**: Future studies should incorporate additional variables, such as speed limits, vehicle types, and driver demographics, to improve analysis depth.
2. **Preventive Road Safety Measures**
    a. **Road Surface Maintenance**: Focus on repairing and improving non-ideal road surfaces, particularly in high-traffic or high-risk areas.
    b. **Seasonal Preparedness**: Implement proactive measures during adverse weather conditions, such as preemptive salting of roads or issuing public warnings.
3. **Technological Enhancements**
    a. **Smart Traffic Systems**: Invest in intelligent traffic management systems that adapt to changing environmental conditions to guide drivers in real time.
    b. **Vehicle Safety Features**: Promote technologies like anti-lock braking systems (ABS) and traction control systems in vehicles to reduce the impact of adverse surface conditions.
4. **Education and Awareness Campaigns**
    a. **Public Outreach**: Develop targeted campaigns to educate drivers about the risks of non-ideal surfaces and encourage safe driving practices in adverse conditions.
    b. **Training Programs**: Offer driver training that emphasizes handling vehicles under challenging environmental scenarios.

11. Takeaways

1. **Environmental Factors Are Context-Dependent**
    a. The interaction between road surfaces and weather conditions plays a more significant role in injury risk than either factor alone.
    b. Adverse road surfaces, regardless of weather, are critical to address for injury reduction.
2. **Focus Areas for Improvement**
    a. Non-ideal road surfaces represent the most actionable area for intervention.
    b. Simple but effective solutions like better road materials or improved drainage systems can significantly reduce crash-related injuries.
3. **Policy Implications**
    a. Road safety interventions should prioritize high-risk combinations of environmental factors, such as adverse weather on non-ideal surfaces.
    b. Encouraging stricter adherence to road maintenance schedules and increasing funding for safety initiatives are necessary steps.

12. Future Discussions/Possibilities

1. Future research can explore the impact of:
    a. **Traffic Density**: Assessing how congestion influences crash severity.
    b. **Vehicle Types**: Identifying whether certain vehicles are more prone to crashes under specific conditions.
    c. **Driver Behaviour**: Investigating human factors like distraction or fatigue.
2. **Advanced Analytical Methods**

Applying predictive modeling techniques (e.g., machine learning) to explore complex interactions between variables and predict high-risk scenarios.

3. **Regional Variations**

Analyzing crash patterns based on geographic and socio-economic factors could provide insights into location-specific safety interventions.

4. **Intervention Impact Studies**

Evaluating the effectiveness of specific interventions, such as road resurfacing or improved lighting, in mitigating crash injuries.

13. References

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